

Mohammed Umair Jibran

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EXPERIENCE

Technology Manager

May 2025 – Present

U+ Education | Progressive Co-op and Contract Roles

York Region, ON

- **Maintained and improved public-facing webpages from stakeholder requirements**, translating requested changes into scoped Jira tickets, implementing and testing updates, and tracking defects, feedback, and approvals from intake through release.
- **Supervise 200+ volunteers and directly manage six volunteer supervisors** across data, software, and web teams, define assignments and timelines, review technical deliverables and dashboards, resolve blockers, and track **2,700+ service hours**.
- Performed peer **QA** and **GitHub code reviews** across frontend interfaces, backend functions, and user forms, identifying functional issues and coordinating corrections with developers.
- **Co-authored a 10-phase delivery plan for a drone training program serving 270+ youth**, establishing scope, schedules, ownership, risk controls, reporting routines, and a **\$35.5K baseline budget**.

Software Developer & Co-Project Coordinator

Feb 2025 – Jan 2026

Islamic Books & Souvenirs | Node.js, JavaScript, Shopify, JSON/CSV

Toronto, ON

- **Built and deployed a Shopify Order Analyzer** that processed JSON/CSV order data, evaluated dimensions against packaging rules, and generated standardized packing instructions, reducing processing time and costs by approximately **15%**.

Technology Lead

Sep 2024 – Apr 2026

Muslim Student Association at York University

Toronto, ON

- **Designed and deployed a WisePOS/BBPOS, Stripe, and JavaScript donation system** supporting more than **\$10,000 in fundraising**, validated terminal hardware and payment data flows, and trained volunteers.

TECHNICAL PROJECTS

CAN-FD Real-Time ECU System | STM32, C/C++, FreeRTOS, CAN-FD, Python

- **Developed and integrated two STM32 ECUs in C/C++ and FreeRTOS**, with a sensor node collecting and transmitting temperature and RPM data while an actuator node validated message identity, integrity, and freshness before controlling a fan.
- Implemented queues, interrupts, timers, and buffers to coordinate real-time CAN-FD message processing while accounting for memory constraints in the embedded system.
- **Created a Python analysis and integration-test tool** to capture CAN-FD traffic, inject delayed or dropped frames, reproduce communication failures, and verify timeout handling, fail-safe transitions, and recovery.

RF Asset Finder | Arduino, C/C++, 433 MHz RF, Custom Packet Protocol

- **Built and tested a low-power RF tag and base station in C/C++**, integrating hardware and software to identify and locate shared equipment through a 433 MHz communications link.
- **Designed and tested a custom packet protocol** with device IDs, CRC validation, acknowledgements, and retransmission logic to reject invalid data and recover from missed responses.

SentriCode – Automated Security Analysis Platform | TypeScript, Python, FastAPI, Docker, PostgreSQL

- **Built a full-stack software tool** that ingests GitHub repositories, coordinates specialized scanners, normalizes findings, calculates risk, and presents results through REST APIs and a web dashboard.
- Designed modular services and shared data contracts for repository profiling, automated analysis, scan-job state, normalized findings, and severity scoring, with Docker-isolated execution and secret redaction.

EDUCATION

York University

Toronto, ON

Bachelor of Engineering in Computer Engineering

Expected 2028

- **Gordon and Agnes (Twambley) Brash Award in Engineering:** Sole engineering student recipient
- **Relevant Coursework:** Embedded Systems, Communication Networks, Electrical Circuits, Electronic Circuits and Devices, Computer Organization, Operating Systems, Advanced Object-Oriented Programming, Database Systems and Design, and Engineering Project Management.

LEADERSHIP & COMMUNITY ENGAGEMENT

Selected Speaker, OACAO Aging Well Conference | Chosen to represent U+ Education as a speaker at the provincial conference
Community Engagement | Represented U+ Education at engagements with MP Tim Hodgson and Richmond Hill Mayor David West

TECHNICAL SKILLS

Programming & Data: Python, SQL, C, C++, Java, JavaScript, PostgreSQL, MySQL, Prisma ORM, JSON/CSV processing
Analytics & Reporting: Power BI, Tableau, Excel, dashboard review, data visualization, data validation, performance analysis
Engineering & Systems: Arduino, CAN-FD, UART, SPI, I2C, 433 MHz RF, hardware-software debugging
Project Delivery & Tools: Requirements, schedules, milestones, budgets, risk and issue tracking, process documentation, technical reporting, Microsoft Office Suite (Excel, Word, PowerPoint, Outlook), Git, GitHub, Jira, Docker